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Subject: $\mathbb{R}$eal Analysis
Class Homework(s)
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Week 1:

1. Each of the four cards shown below has a number on one side and a letter on the other side. What is the minimum number of cards that must be turned in order to prove the correctness of this statement: "If a card has a vowel on one side, then it has an even number on the other side?"

$$\boxed{A}\boxed{C}\boxed{4}\boxed{7}$$

Proof. We only need to check card $\boxed{A}$ and $\boxed{7}$. We check $\boxed{A}$ because if that is false, the claim doesn't hold. Also note the contrapositive of *If one side has vowel then there's even number on the other side* will be *If one side has odd number then there's a consonant on the other side.* So we check $\boxed{7}$ as well. The statement says nothing when one side is a consonant so we ignore that. The $\boxed{4}$ card will work either way so also ignored. □

2. Is there a explicit bijection from $\mathbb{N} \to \mathbb{Q}$ or other way around?

Proof. Yes. Consider $S : \mathbb{N} \to \mathbb{Q}$ such that

$$S(x) = \begin{cases} \frac{\delta(x)}{2\lfloor x \rfloor - x + 1} \\ 0, x = 0 \end{cases}$$

The sequence $S(0), S(S(0)), \cdots$ gives all positive rationals uniquely. □

Week 2: Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Prove that if f is bounded on every closed interval, then f is bounded on $\mathbb{R}$.

Week 3: Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Prove that if f is bounded on every closed interval, then f is bounded on $\mathbb{R}$.